WEEK 2 – DATA SCIENCE AND VISUALIZATION

Dataset Representation and Initial Exploration

Project: Spotify Songs Analysis

1. Introduction

This week's work focuses on the initial representation and exploration of the Spotify Songs dataset as part of the Data Science and Visualization (DSV) project. The purpose of this stage is to understand the structure, characteristics, and contents of the dataset before performing data cleaning and exploratory data analysis. The dataset contains information about Spotify tracks, albums, artists, and various audio features.

2. Dataset Description

The project uses the Spotify Songs dataset, containing approximately 19,335 song records and multiple attributes describing each track. The dataset includes track and album information such as track ID, track name, artist information, album information, release date, track number, album type, Spotify URLs and identifiers, and explicit content information. It also contains audio features including danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration, and time signature.

3. Objectives

The main objectives of Week 2 were to load and understand the Spotify dataset, inspect its structure and dimensions, identify the available attributes, distinguish numerical and categorical attributes, examine basic statistical characteristics of numerical variables, represent the dataset in a structured manner, and prepare it for subsequent data-analysis stages.

4. Tools and Technologies Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, and Seaborn were used for dataset representation and initial analysis.

5. Dataset Loading

The dataset was loaded into a Pandas DataFrame. Each row represents a song, while each column represents an attribute or feature of that song.

6. Dataset Representation

The first records of the dataset were examined using `df.head()`, while `df.tail()` can be used to inspect the final records. The dimensions were examined using `df.shape`. The dataset contains approximately 19,335 records and multiple track, album, and audio-feature attributes.

7. Dataset Structure

The structure and data types of the columns were examined using `df.info()`. This identifies numerical columns, categorical/text columns, missing values, and memory usage. The dataset contains both categorical and numerical attributes. Important numerical audio features include danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration, and time signature. Categorical attributes include artists, track name, album name, album type, explicit content, release-date information, and Spotify identifiers/URLs.

8. Statistical Representation

Basic statistical characteristics of numerical features were examined using `df.describe()`. This provides count, mean, standard deviation, minimum, quartiles, and maximum values. Initial statistics indicate average danceability of approximately 0.616, energy of 0.584, acousticness of 0.309, valence of 0.491, tempo of 119.79 BPM, and loudness of -8.60 dB.

9. Data Representation Through Visualization

Visualization provides an intuitive understanding of the dataset. Histograms and KDE plots can be used to represent distributions of important audio features such as danceability, energy, tempo, loudness, acousticness, and valence. These visualizations help identify concentration of values, skewness, and unusual observations.

10. Initial Observations

The dataset contains a large number of Spotify tracks and is suitable for exploratory data analysis. It contains both categorical and numerical attributes. The audio features provide quantitative measurements that can be analyzed statistically. Different audio features have different value ranges and scales. Some metadata fields contain missing values and require further investigation. Features such as danceability, energy, acousticness, and valence are generally represented within a normalized range, while tempo is measured in beats per minute and loudness uses a different numerical scale.

11. Preparation for Further Analysis

The dataset representation performed during Week 2 establishes the foundation for subsequent stages of the Spotify DSV project. The dataset is now ready for detailed data-cleaning and exploratory-analysis activities.

12. Conclusion

Week 2 focused on understanding and representing the Spotify Songs dataset before applying advanced data-processing techniques. The dataset was loaded and examined using Python and Pandas. Its structure, dimensions, data types, numerical features, categorical features, and basic statistical characteristics were studied. This initial representation established a clear understanding of the dataset and prepared it for subsequent analysis.

13. Technologies Used

Technology	Purpose
Python	Programming and analysis
Pandas	Data loading and manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook	Interactive analysis environment
GitHub	Project version control and documentation

14. Project Status

Week 2: Dataset Representation and Initial Exploration — Completed